Data Structures and Algorithms Project
TicTacToe game with AI

DUE DATE: Friday 3rd APRIL 2026

This Project is worth 15%

Project Group Size: 4 to 5 persons

NOTE: Students can form groups from members from any other lab session.

1 Overview

1.1 Project Steps

Tic-Tac-Toe (Classic) Game:
Tic-Tac-Toe is a classic two-player strategy game usually played on a 3×3 grid. Each player is assigned a symbol—traditionally X or O—and takes turns placing their symbol in an empty square on the grid. The objective is to be the first player to arrange three of their symbols in a continuous line—horizontally, vertically, or diagonally. If all squares on the board are filled and neither player has formed a line of three, the game ends in a draw. **This project will use a standard 3 × 3 grid.**

A detailed overview of the Tic Tac Toe (classic) game can be found here:
https://www.youtube.com/watch?v=3qzcAMShotQ

Your group must create a fully working intelligent Tic-Tac-Toe command line game in Java using only basic arrays (e.g., char[][] or int[][]). **Higher-level data structures beyond simple arrays will result in a maximum achievable mark of 50% for this assignment**. The game must allow the user to select the mode of the game. The two modes are:

- 2 player (2 human players)
- 1 player (Human vs Intelligent AI)

Intelligent AI:
This must be implemented using the **Minimax algorithm**. The computer must examine the game state at each point and deduce the optimal (best) play that can be made.

A detailed overview of the Minimax algorithm can be found here:
https://www.youtube.com/watch?v=zp3VMe0Jpf8

<u>The Game should operate as follows:</u>

1) The user should be asked if it is a 2 player OR 1 player game (versus Intelligent AI)
2) If the user selects **1 player (versus minimax AI)**
    a. The Human player should be asked their name
    b. The player must be prompted to choose their symbol ('**X**' or '**O**')
    c. The player with symbol '**X**' should go first (computer or human player).
    d. The present board state will be displayed and:
        1. If the computer is to play, it will now show the board updated with the computer's symbol.
        2. If it is the player's turn, the player will be prompted for a row and column location to play.
    e. The board will then be checked for a winning state or draw and if it is not in a winning state or draw(tie), go to **<u>step d</u>** otherwise display the winner or declare a draw (tie).

3) If the user selects **2 player (2 Human players)**
    a. Each human player must be prompted to choose their name and symbol ('**X**' or '**O**')
    b. An empty board will then be displayed.
    c. The next player to move will be prompted for a row and column location to play.
    d. The present board state will be displayed showing their updated play.
    e. The board will then be checked for a winning state or draw and if it is not in a winning state or draw, go to **<u>step c</u>** otherwise display the winner or declare a draw.

At the end of a game, the result should be displayed.

<u>YOU MUST ONLY USE BASIC ARRAYS TO MODEL THE BOARD STATE – if this is not done, the maximum mark that can be attained is 7.5/15.</u>

NOTE:

The non-AI part (2 human players) is worth <u>10 of the marks</u>

The recursive minimax version (1 human vs AI) is worth <u>5 of the marks</u>

This assignment loses 30% each day (based on the date) after the 3rd April.

- **This means if it is submitted 12:01 AM (1 minute late) on the 4th, the maximum mark that can be got is 10.5/15.**
- **This means if it is submitted 12:01 AM on the 5th, the maximum mark that can be got is 6/15.**

It is therefore better to submit a non-working AI version on time than an AI version late.

1.2 Submission Format

This is a group assignment, and one report and code will be submitted per group.

Your group must submit two Items:

1) The completed submission document (Given with this project) **with ALL the group member's names** inserted at the top and code pasted neatly into it.

2) The completed Java source classes (zipped) with all student's names and numbers commented at the top of the code [not the whole project, just all the java classes you created]. **To be clear, do NOT upload a whole solution just the classes**.

Please note:

1) 2 marks will be automatically deducted if the submission formats are not followed.